VENKATESH NANDIKOLLA

SENIOR DATA ENGINEER | CLOUD DATA PLATFORMS

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7.9 years of experience delivering data platforms, batch and streaming pipelines, and reporting solutions across AWS and Azure. Strong hands-on experience with Snowflake, Databricks and Amazon Redshift, supported by SQL, Python, PySpark and PostgreSQL. Expertise spans ETL/ELT, dimensional modeling, CDC, data quality and performance optimization. Delivered production data pipelines and cloud warehouse migrations, supporting analytics, billing and financial-services reporting in collaboration with cross-functional teams.

TECHNICAL SKILLS

Databases and SQL: Snowflake, Amazon Redshift, Aurora PostgreSQL, Oracle; SQL, stored procedures, PL/pgSQL

Snowflake: Snowpipe, S3 and SQS integration, COPY INTO, clustering, virtual warehouses, performance optimization, grants and role-based access control (RBAC)

Data engineering: Batch and streaming pipelines, ETL/ELT, dimensional modeling, medallion architecture, SCD Type 1/2, CDC, data quality, reconciliation; dbt (POC)

Cloud and processing: AWS S3, SQS, MSK, Redshift; Azure Databricks, ADLS Gen2; Python, Spark, PySpark, Delta Lake, Kafka, Flink, Hadoop, Hive

Delivery and operations: SQL performance tuning, Redshift workload management (WLM), Git, GitHub, Jenkins, Jira, Agile/Scrum

AI and automation: Claude Code, Claude in VS Code, MCP servers, AI-assisted workflow automation; knowledge of agents and skills; project experience with OpenAI embeddings and retrieval-augmented generation (RAG)

PROFESSIONAL EXPERIENCE

SS&C Technologies | Senior Data Engineer

Jan 2024 - Oct 2026 | Hyderabad

- Designed Redshift SQL processing for Oracle CDC events ingested through Amazon MSK and streaming materialized views; developed stored procedures scheduled every five minutes for current-state and historical reporting datasets.
- Implemented CDC versioning with business keys, source timestamps, event metadata and SQL window functions; handled hard deletes, late-arriving records and duplicate replays through SCD Type 1/2 logic and reporting views.
- Developed dimensional models and PostgreSQL reporting functions for analytics and billing; tuned queries using pre-aggregation, temporary-table indexing and ANALYZE.
- Built Oracle-to-Redshift reconciliation using record counts and business-column hashes to validate migration results and investigate data discrepancies.
- Managed Redshift access grants and used WLM to separate workloads and prioritize query execution; implemented report-level access controls for PostgreSQL reporting.
- Evaluated dbt in a proof of concept for SQL transformations; used GitHub, Jira and Scrum practices for collaborative delivery.
- Implemented Flink transformations for microservice events through Interconnect, a RabbitMQ connector and MSK; maintained Aurora PostgreSQL tables with timestamp-ordered upserts that reject stale or duplicate events.
- Led Oracle-to-microservices migration workstreams, integrating historical data and ongoing events for reporting continuity; supported Aurora analytics through Redshift federated queries and zero-ETL replication.
- Built AI-assisted migration orchestration using Claude in VS Code, MCP and Jenkins across Oracle-to-Redshift loading, transformations, PostgreSQL processing and data-quality checks.

Ernst & Young (EY) | Senior Consultant - Data Engineer

Aug 2021 - Dec 2023 | Hyderabad

- Built production Snowflake ingestion using Amazon S3, SQS notifications and Snowpipe to automate loading of incoming data into the warehouse.
- Developed Snowflake SQL stored procedures and transformations, and designed dimensional models to organize data for reporting and analytics.
- Optimized Snowflake performance through clustering and separate virtual warehouses for different workloads; managed database access through grants and role-based permissions.
- Developed production Azure Databricks, PySpark and Delta Lake pipelines processing approximately 40 GB/day across CSV, JSON, Parquet, XLSX and compressed files.
- Implemented medallion data layers for ingestion, cleansing and curated consumption, using a unified data model and validation controls to improve consistency across pipeline stages.
- Supported technical delivery, code reviews and coordination across an eight-member project team; used GitHub for version control and Jira within Agile/Scrum delivery.

Cognizant Technology Solutions | Programmer Analyst - Big Data Developer

Feb 2019 - Jul 2021 | Bengaluru

- Contributed to a legacy data warehouse migration to Snowflake, handling the loading of migration data from Amazon S3 into Snowflake using COPY INTO.
- Developed data validation scripts to check migrated datasets and support source-to-target consistency during platform modernization.
- Built distributed Spark, Hadoop and Hive pipelines for legacy data processing and modernization; converted CSV datasets to Parquet for more efficient storage.

EDUCATION

Bachelor of Technology | Jawaharlal Nehru Technological University, Kakinada

2014 - 2018